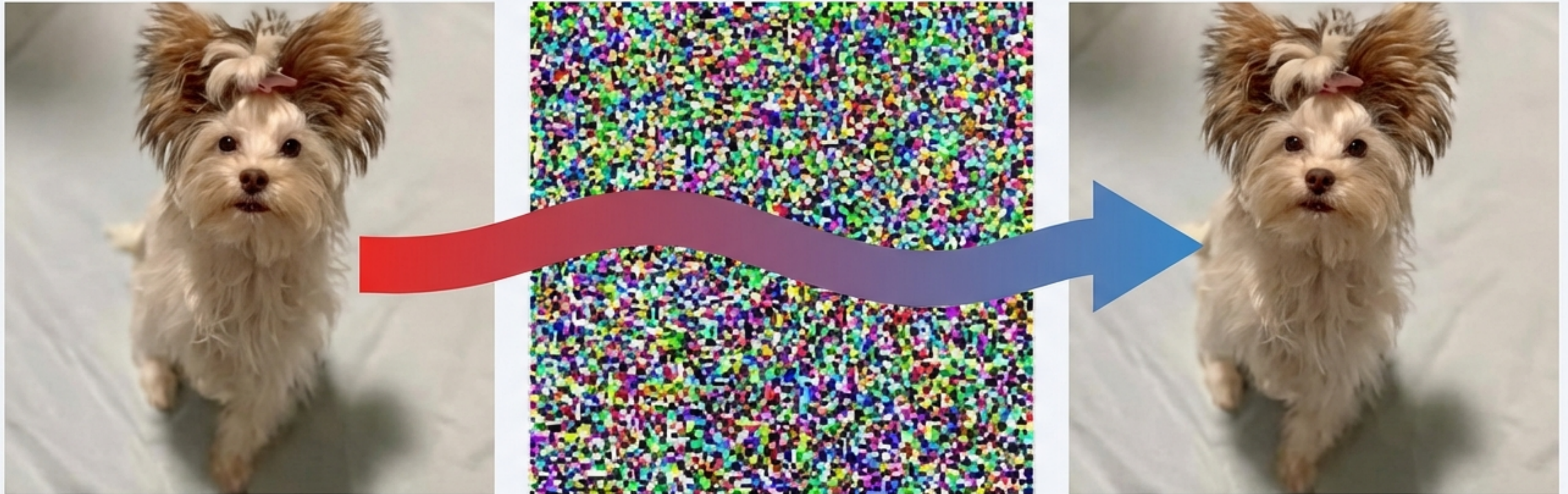


Structure from Chaos: The Science of Diffusion Models

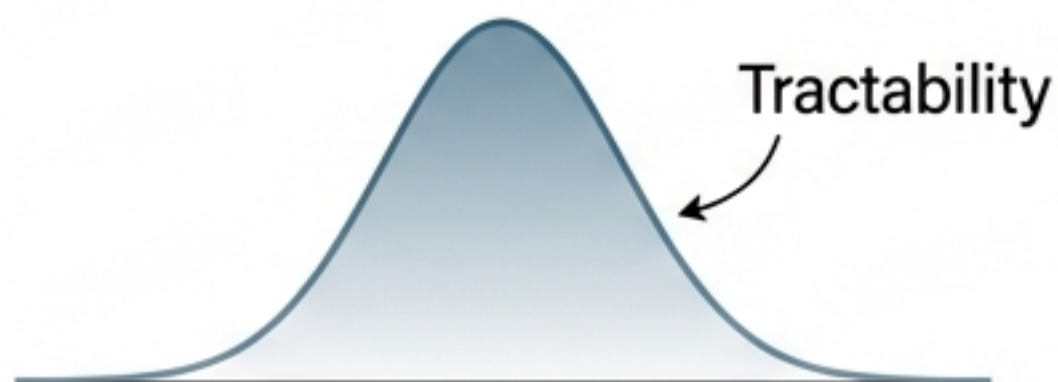
Leveraging Non-equilibrium Thermodynamics for State-of-the-Art Generative AI



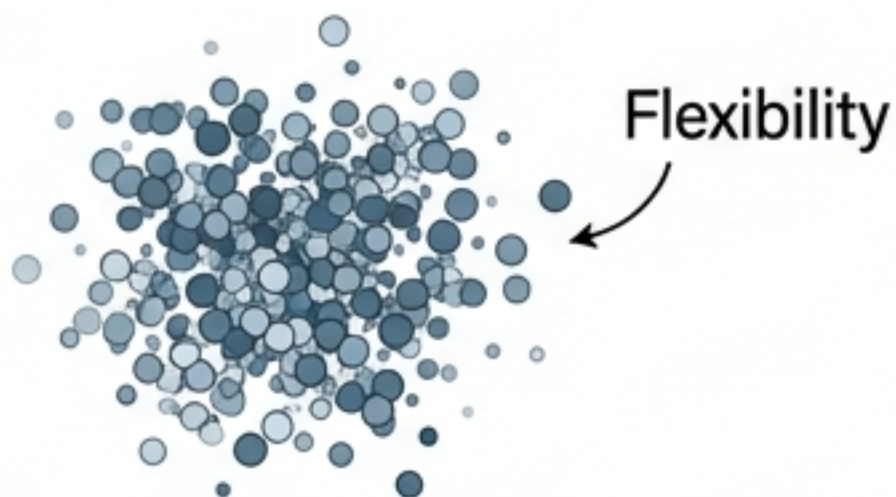
“A central problem in machine learning involves modeling complex data-sets... [Diffusion] simultaneously achieves both flexibility and tractability.” — Sohl-Dickstein et al. (2015)

The Historical Trade-off: Flexibility vs. Tractability

The Generative Dilemma



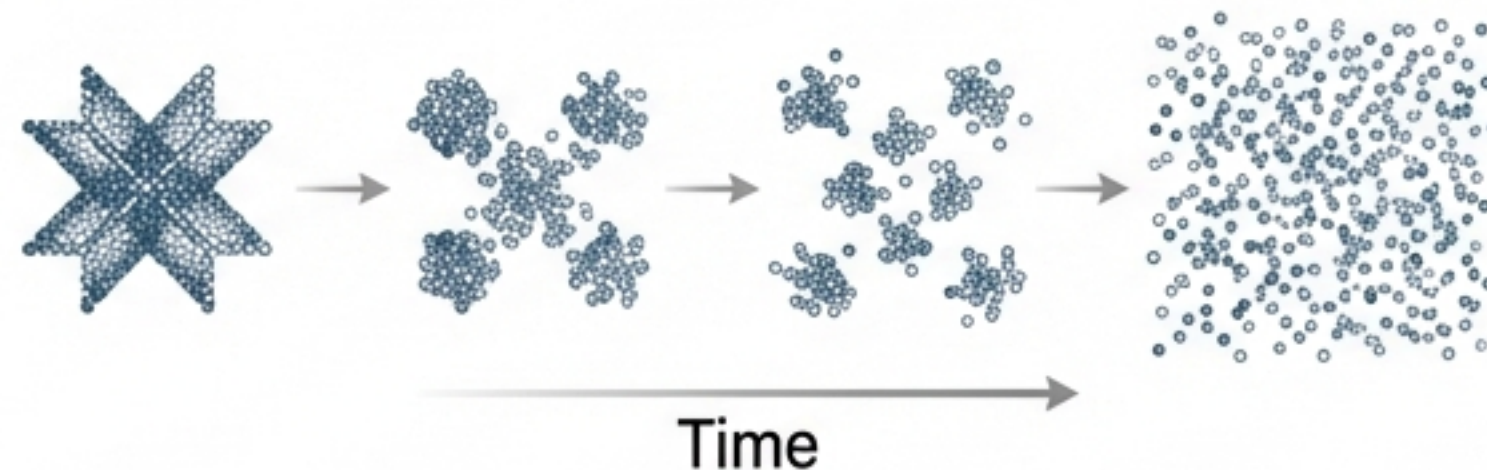
Simple models (e.g., Gaussians) are easy to evaluate but fail to capture rich data structures.



Complex models fit data well but require intractable normalization constants or expensive Monte Carlo sampling.

The Thermodynamic Insight

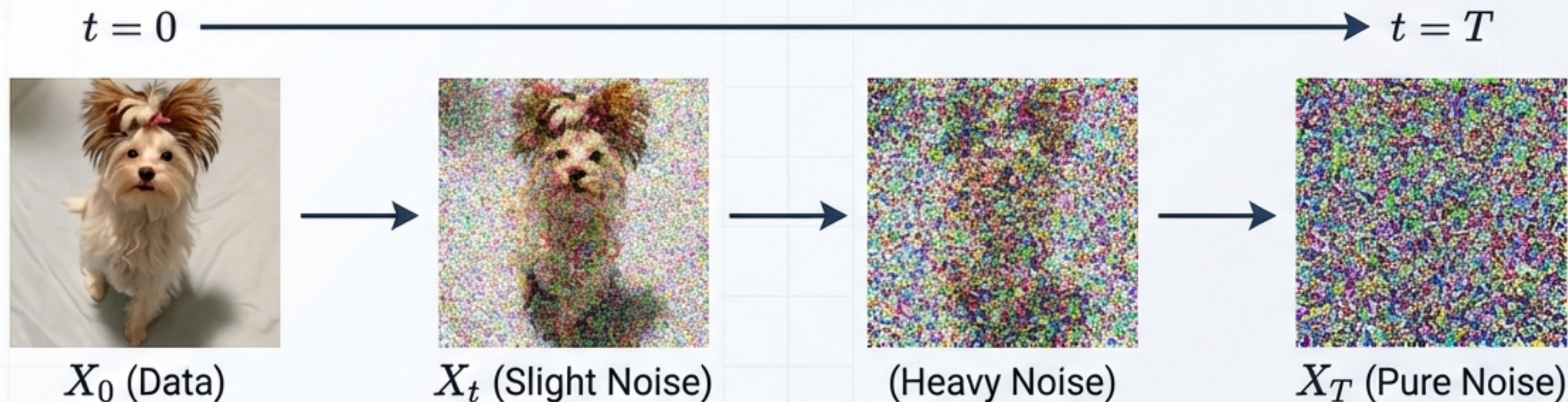
The Solution: Instead of modeling complex data all at once, we slowly destroy it.



Systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process

We trade the difficulty of modeling complex data for the tractability of modeling a simple decay process.

The Forward Process: The Road to Entropy

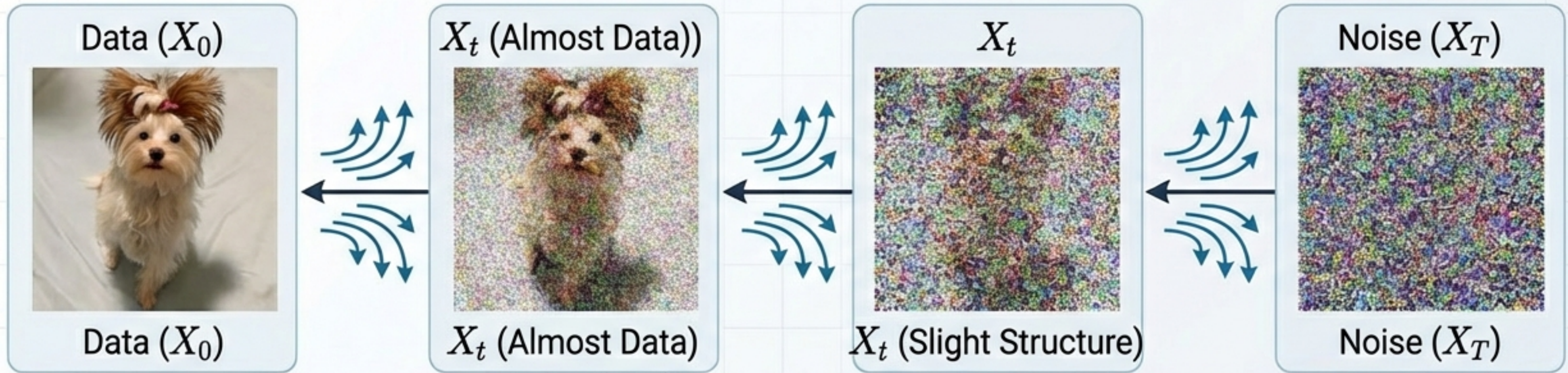


$$dX_t = -\frac{1}{2}g(t)X_tdt + \sqrt{g(t)}dW_t$$

Current State $\nearrow$ Noise Schedule (Corruption Speed) $\nearrow$ Injection of Gaussian Noise (Brownian Motion)

In the infinite-time limit, the complex data distribution is transformed into a simple standard Gaussian distribution $N(0, I)$.

Reversing Time: The Backward Process



$$dX_t^{\leftarrow} = \left[\frac{1}{2} X_t^{\leftarrow} + \boxed{\nabla \log p_t(X_t^{\leftarrow})} \right] dt + dW_t$$

The Score Function

- The gradient of the log probability density.
- Acts as a compass pointing 'uphill' towards higher data density.
- **Visual Metaphor:** A vector field guiding us out of the noise.

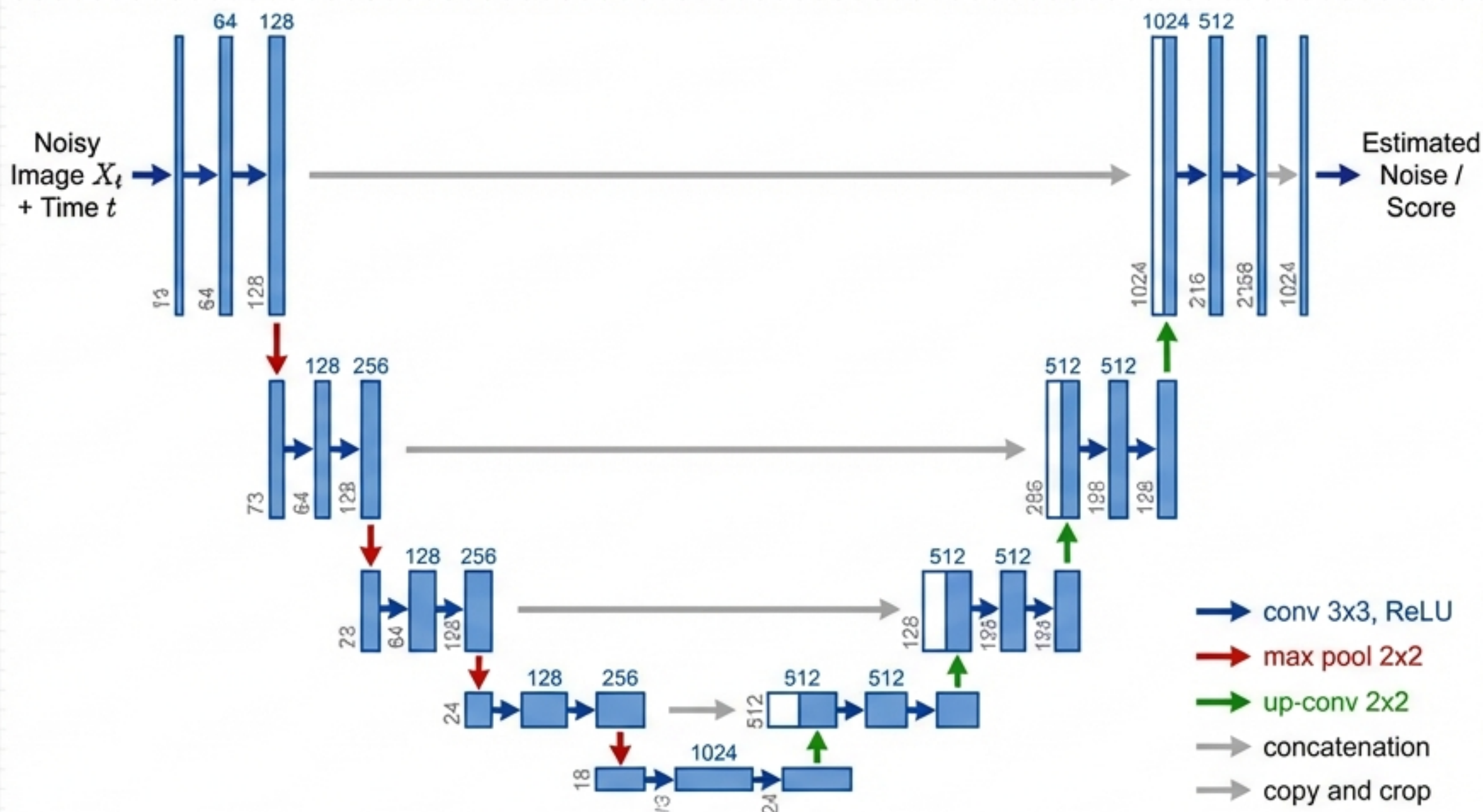
Training the Model: Denoising Score Matching

Concept:

We do not need to know the full probability distribution. We only need to learn to predict the noise added at each step.

Mechanism:

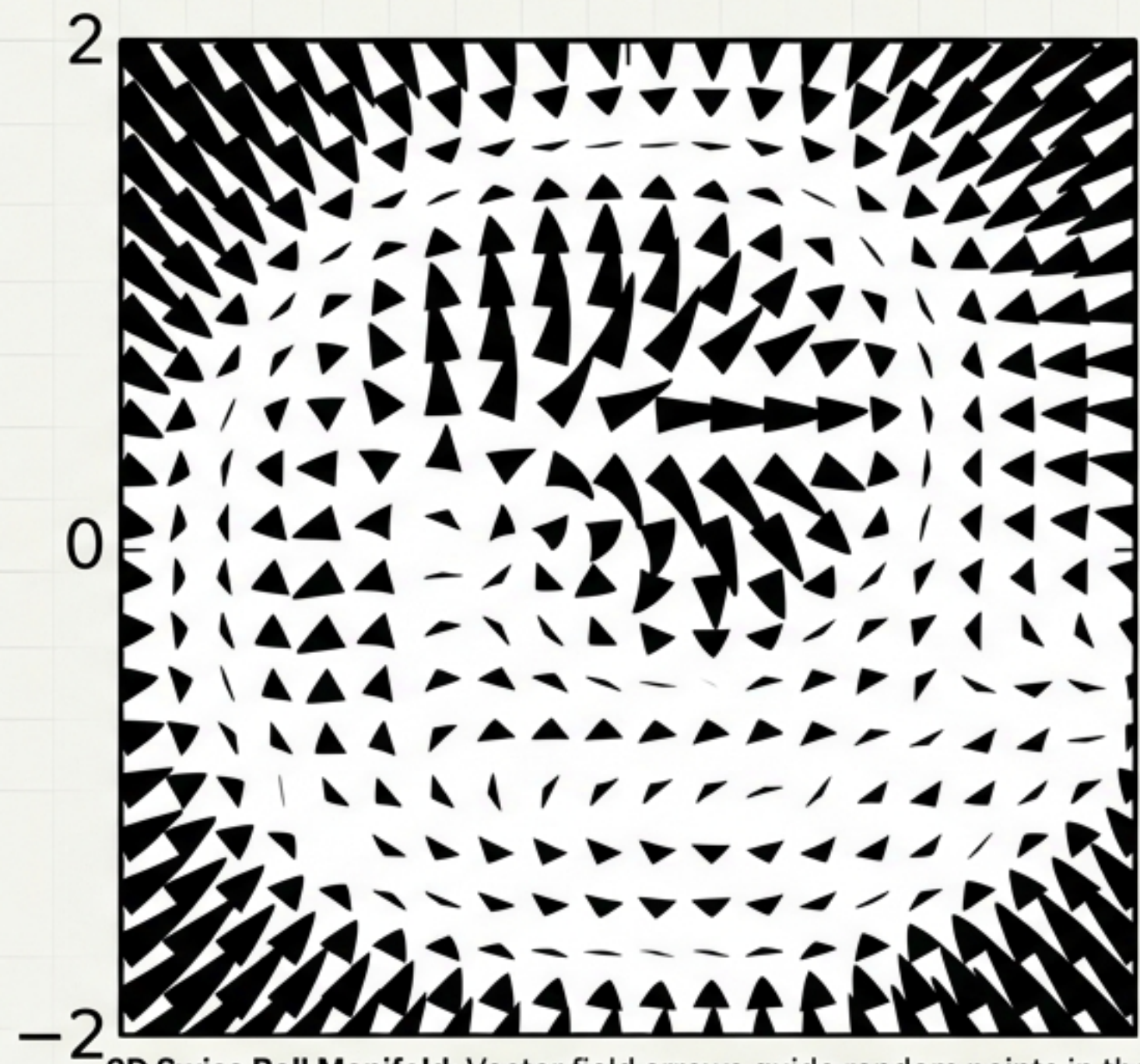
A neural network minimizes the difference between actual noise (known during training) and predicted score.



$$\min_{s \in \mathcal{S}} \mathbb{E} \left[\|\nabla \log \phi_t(x_t | x_0) - s(x_t, t)\|^2 \right]$$

Navigating High Dimensions: The Manifold Hypothesis

- **The Curse of Dimensionality:** Searching the entire space of possible pixel combinations is impossible.
- **The Solution:** Real-world data lives on low-dimensional structures called manifolds.
- The model learns the vector field that collapses random points back onto this structured manifold.



2D Swiss Roll Manifold: Vector field arrows guide random points in the ambient space back towards the data distribution (the spiral).

“The **Score Function** circumvents the dependence on data **ambient dimension**... identifying the **geometry of the data subspace**.” — Chen et al. (2024)

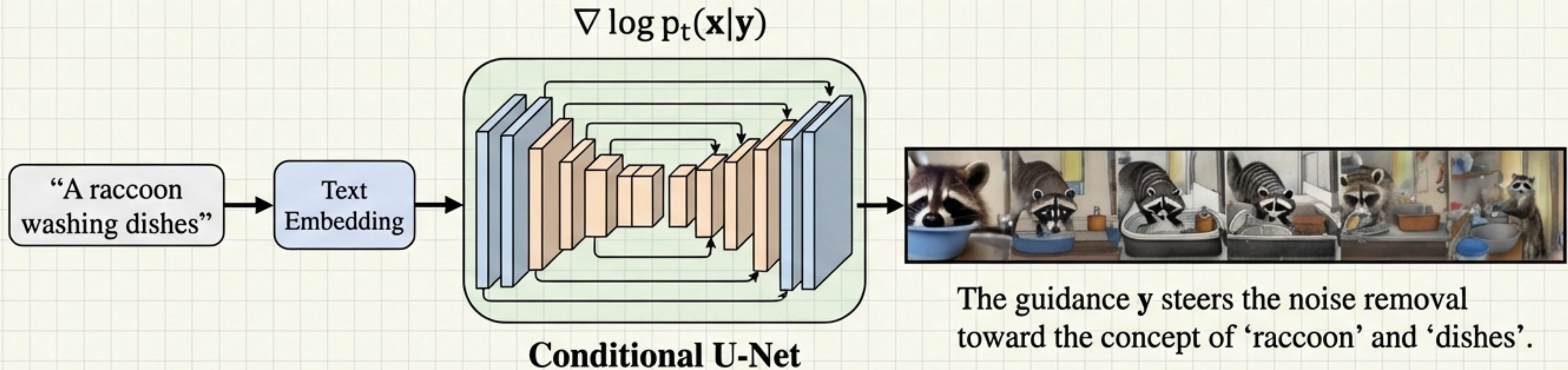
Theoretical Guarantees: Why It Converges



- **Approximation:** If the Score Function is accurately estimated ($L2$ error is small)...
- **Sampling:** ...the generated distribution tracks the real data distribution.
- **Caveat:** The "Score Blowup" issue near $t = 0$ (clean data) requires Early Stopping or specific weighting to maintain stability.

From Randomness to Control: Conditional Diffusion

Unconditional models generate random samples. **Conditional models** generate samples from a specific slice of the manifold guided by external information $\mathbf{y}$. $\nabla \log p_t(\mathbf{x}|\mathbf{y})$



Mechanisms of Guidance

Guidance effectively ‘tilts’ the vector field. The compass now points to ‘valid images that look like y ’.

- **Classifier Guidance:** Uses a pre-trained classifier to calculate gradients $\nabla \log p(y|x_t)$.
- **Classifier-Free Guidance:** Trains a single network to interpolate between conditioned and unconditioned scores.

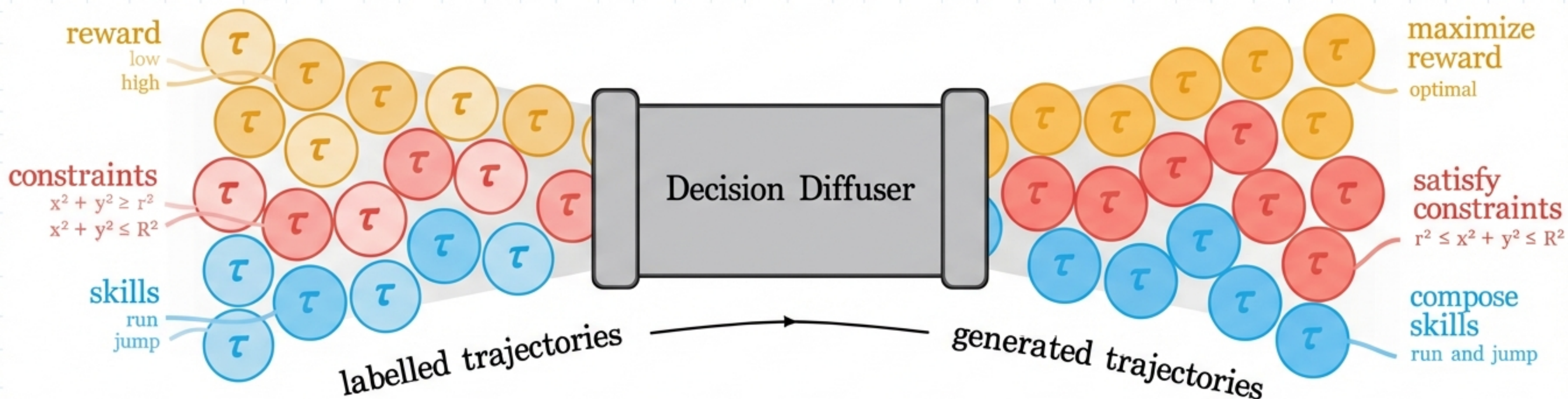


From realistic to stylized based on text prompt guidance.

Emerging Applications: Planning as Generative Modeling

Problem: Traditional Reinforcement Learning maximizes rewards step-by-step.

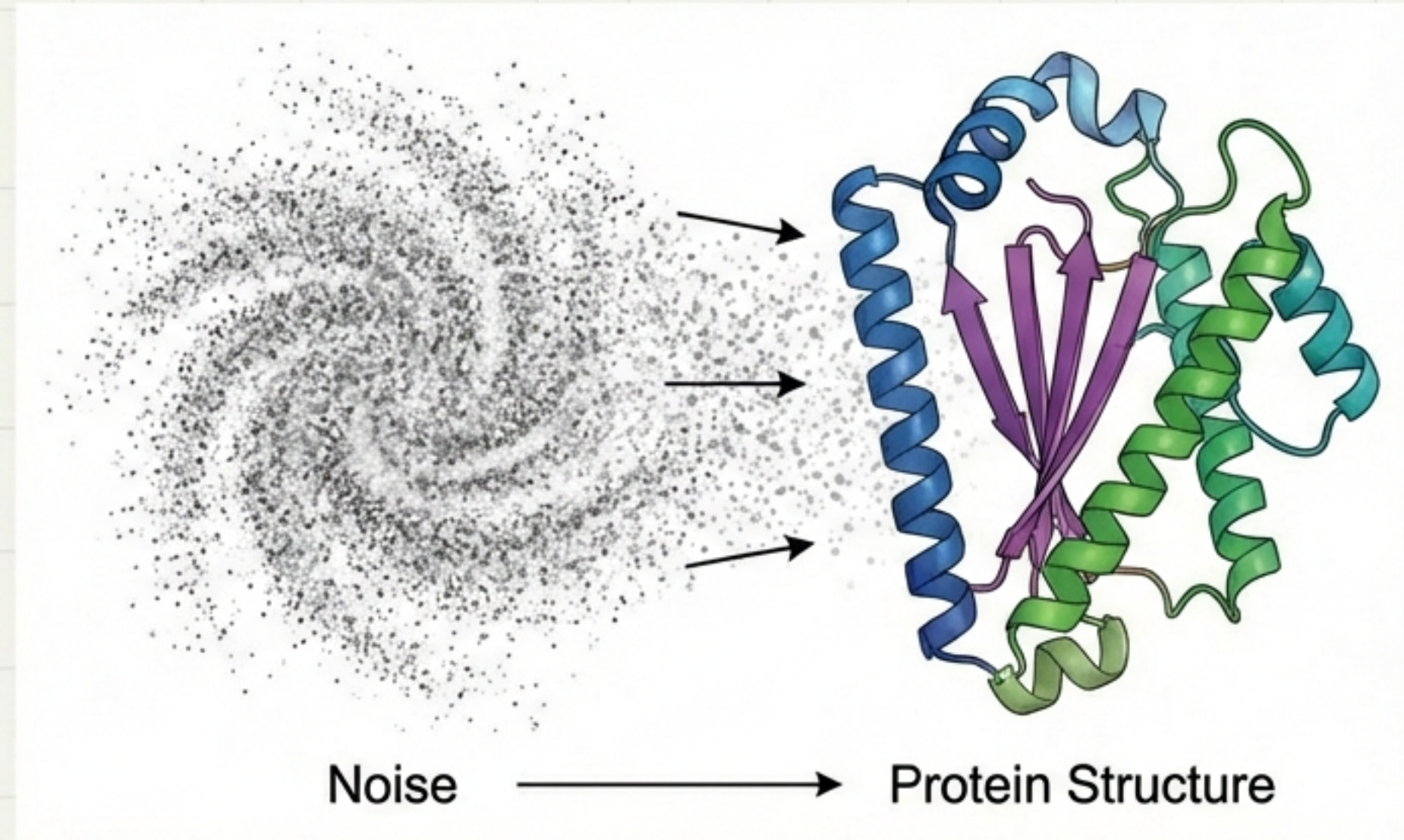
Solution: Diffusion treats a sequence of actions as a single “image” (trajectory) to be generated.



We generate a full state-action trajectory τ conditioned on high reward: $P(\tau \mid \text{Reward} = \text{High})$.

Life Sciences: Modeling the Building Blocks of Life

- Biological data (protein sequences, drug molecules) is high-dimensional but highly structured.
- **Protein Design:** Diffusing noise into a sequence w that folds into a valid structure.
- **Why Diffusion?** It captures the 'manifold of plausible biological structures' better than VAEs or GANs.

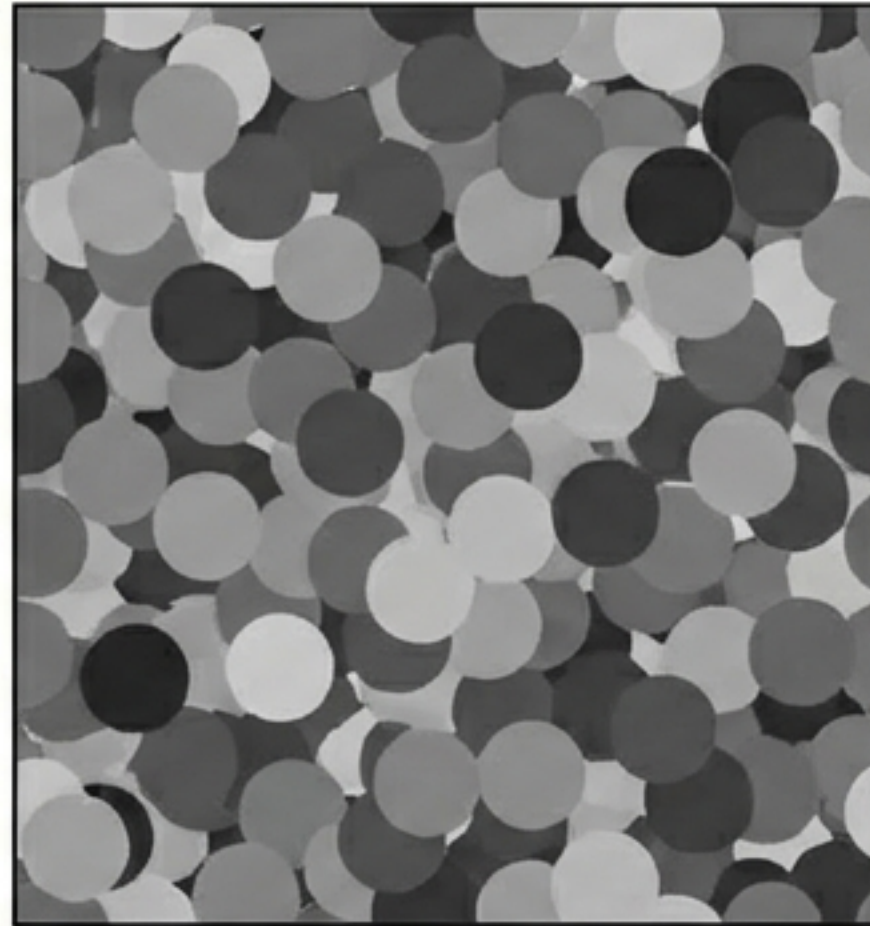


Diffusing noise to structure: Emergence of a functional protein

Optimization as Conditional Sampling

A Paradigm Shift in Optimization:

- **Old Way:** Gradient ascent on an objective function $V(\tau)$. Prone to local optima.
- **Diffusion Way:** Sample from the distribution $P(\tau \mid V(\tau) > a)$.
- **Benefit:** Generates diverse, high-fidelity solutions.



Synthetic Dead Leaves



Real Dead Leaves

Diffusion captures complex, non-smooth structures (like the occlusion of leaves) that traditional optimization fails to model.

The Frontier: Speed, Discrete Structures, and Robustness

Faster Sampling



Simulating 1000+ timesteps is slow. New methods use Distillation, Consistency Models, and ODE solvers to reduce steps drastically.

Discrete Diffusion



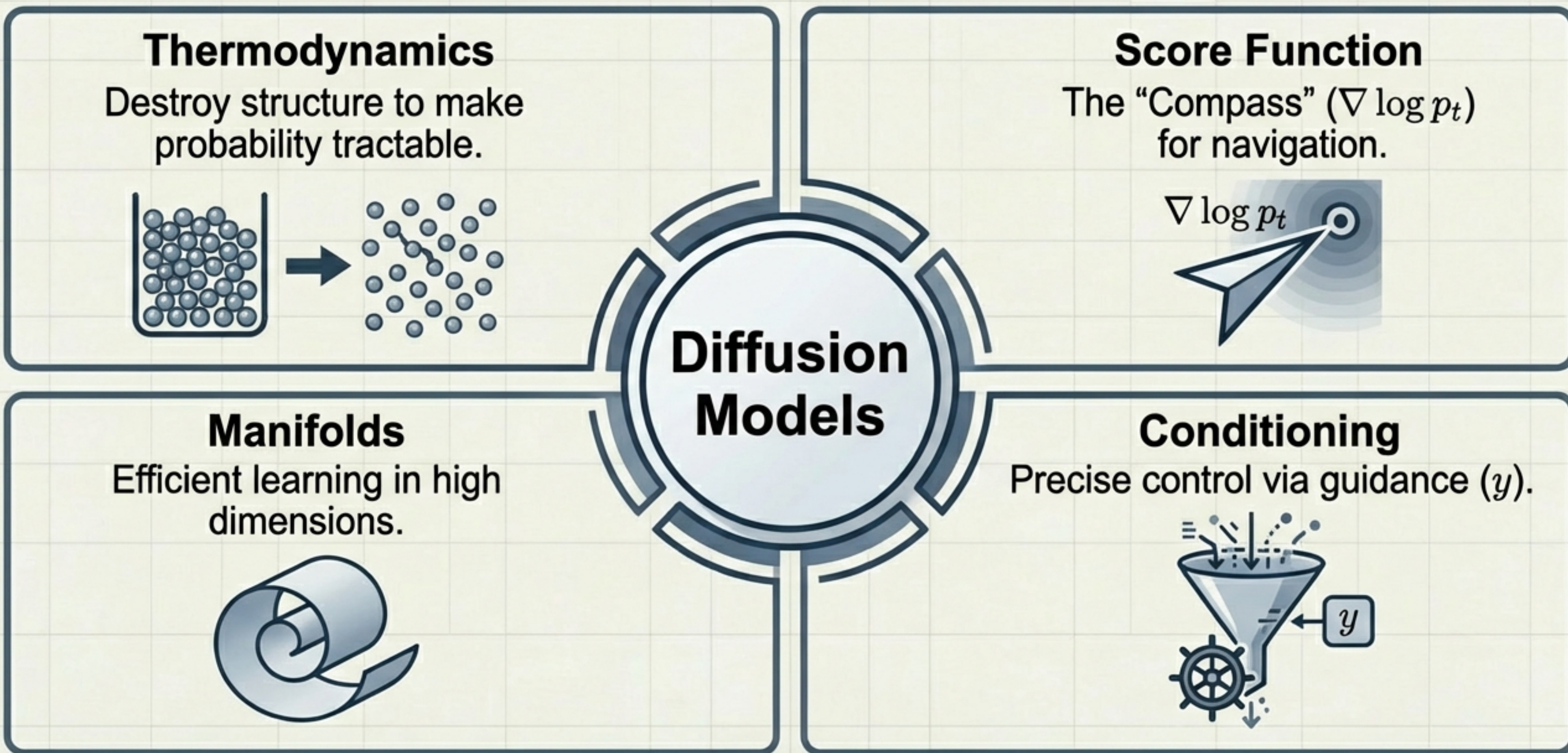
Applying diffusion to non-continuous data like text, graphs, and DNA sequences (Discrete State Spaces).

Adversarial Robustness



Using the generative priors of diffusion models to purify data and defend against adversarial attacks.

Summary: The Power of Controlled Chaos



References & Further Reading

An Overview of Diffusion Models: Applications, Guided Generation, Statistical Rates and Optimization

Authors: Chen, Mei, Fan, Wang (2024).



Scan for Paper

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

Authors: Sohl-Dickstein, Weiss, Maheswaranathan, Ganguli (2015).



Scan for Paper

“A central problem in machine learning involves modeling complex data-sets... [Diffusion] simultaneously achieves both flexibility and tractability.” — Sohl-Dickstein et al.